



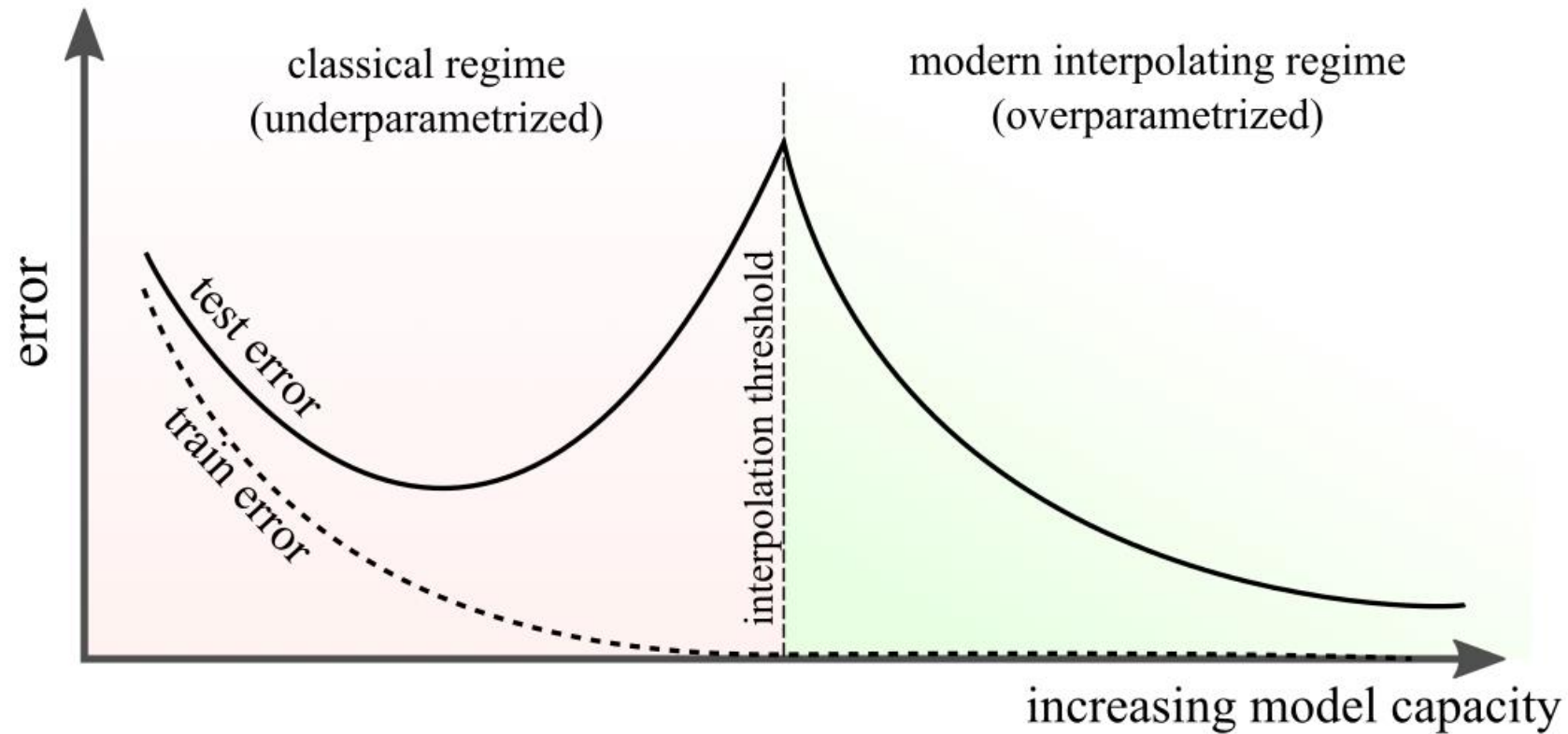
Model Selection for Overparameterized Models

Maurizio Filippone
Statistics Program, KAUST

Double Descent and Neural Scaling Laws



- **Double descent** challenged common wisdom on overfitting!



Double Descent and Neural Scaling Laws



- **Neural scaling laws** allow one to extrapolate generalization wrt to model size, compute, and data set size.

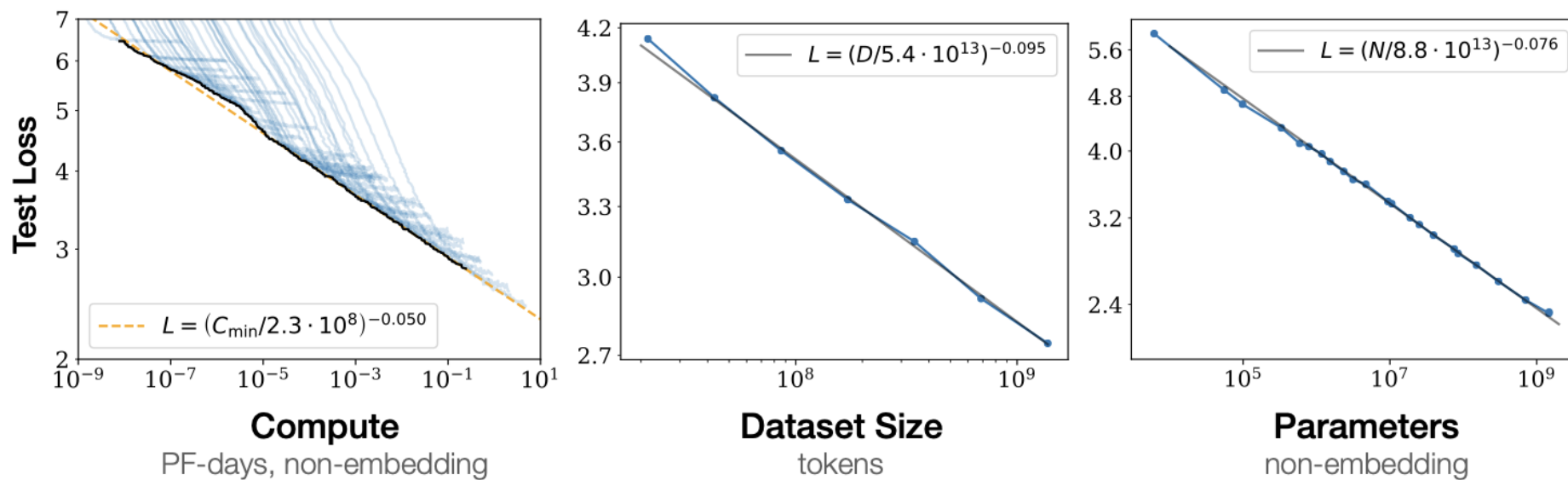


Figure 1 Language modeling performance improves smoothly as we increase the model size, dataset size, and amount of compute² used for training. For optimal performance all three factors must be scaled up in tandem. Empirical performance has a power-law relationship with each individual factor when not bottlenecked by the other two.

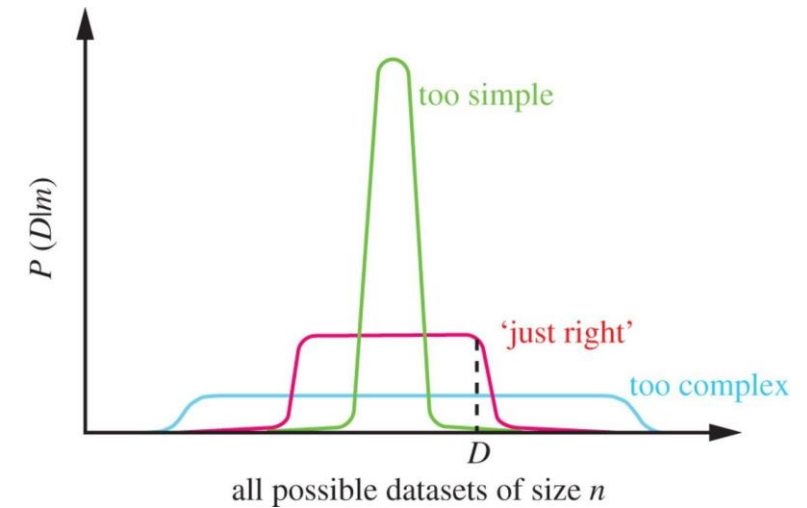
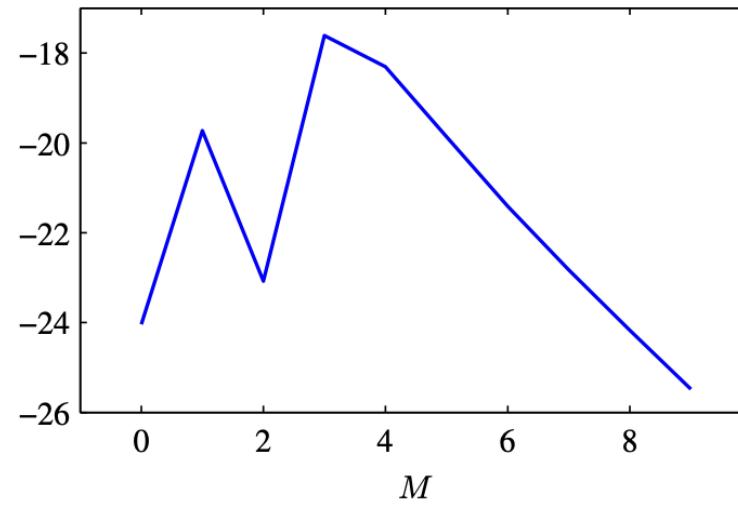
Bayesian Model Selection



- Marginal Likelihood as a model selection criterion.

$$p(\mathbf{W} | \mathbf{Y}, \mathbf{X}) = \frac{p(\mathbf{Y} | \mathbf{X}, \mathbf{W})p(\mathbf{W})}{\int p(\mathbf{Y} | \mathbf{X}, \mathbf{W})p(\mathbf{W}) d\mathbf{W}}$$

Plot of the model evidence versus the order M , for the polynomial regression model, showing that the evidence favours the model with $M = 3$.



Bayesian Model Selection



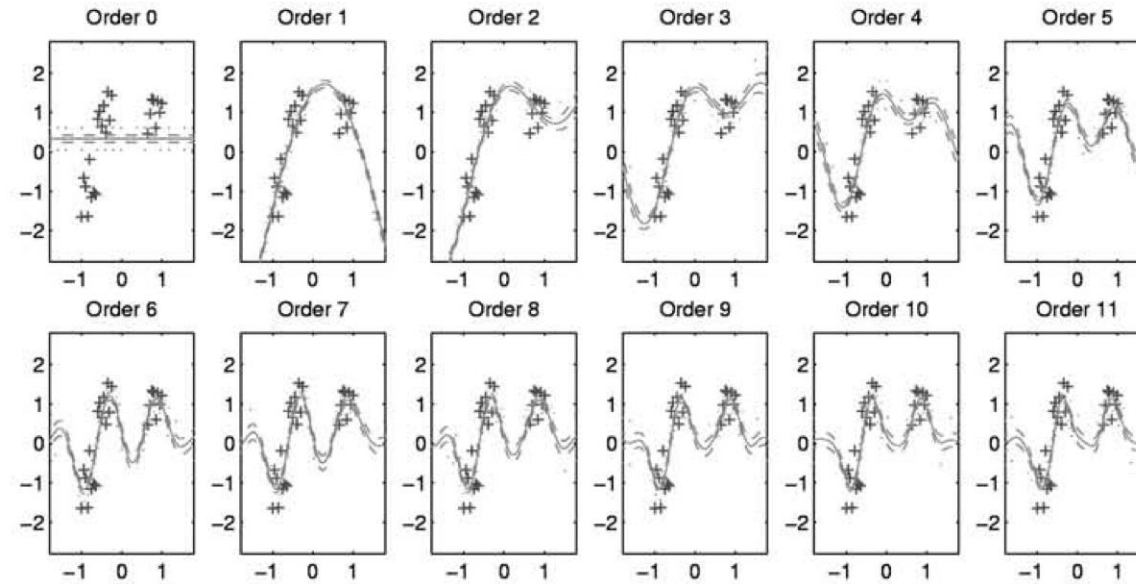
- **Bayesian Model selection** is connected with:
 - **Occam's Razor**: simplest explanation is usually the best one
 - **Kolmogorov complexity**: length of a shortest computer program generating the weights of the network
 - **Minimum Description Length**: find the most concise representation of the data

- How do we reconcile Bayesian model selection with the conclusions from research on scaling laws?

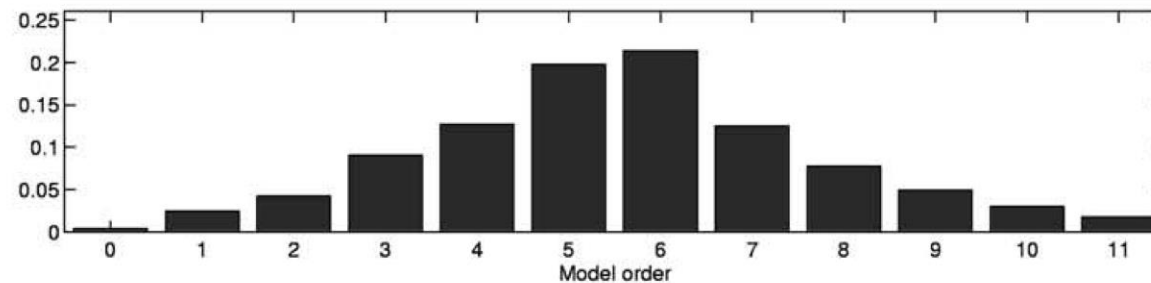
Penalizing Complexity



- Consider **trigonometric regression** as follows:
- $$y(x) = a_0 + \sum_{d=1}^D a_d \sin(dx) + b_d \cos(dx),$$



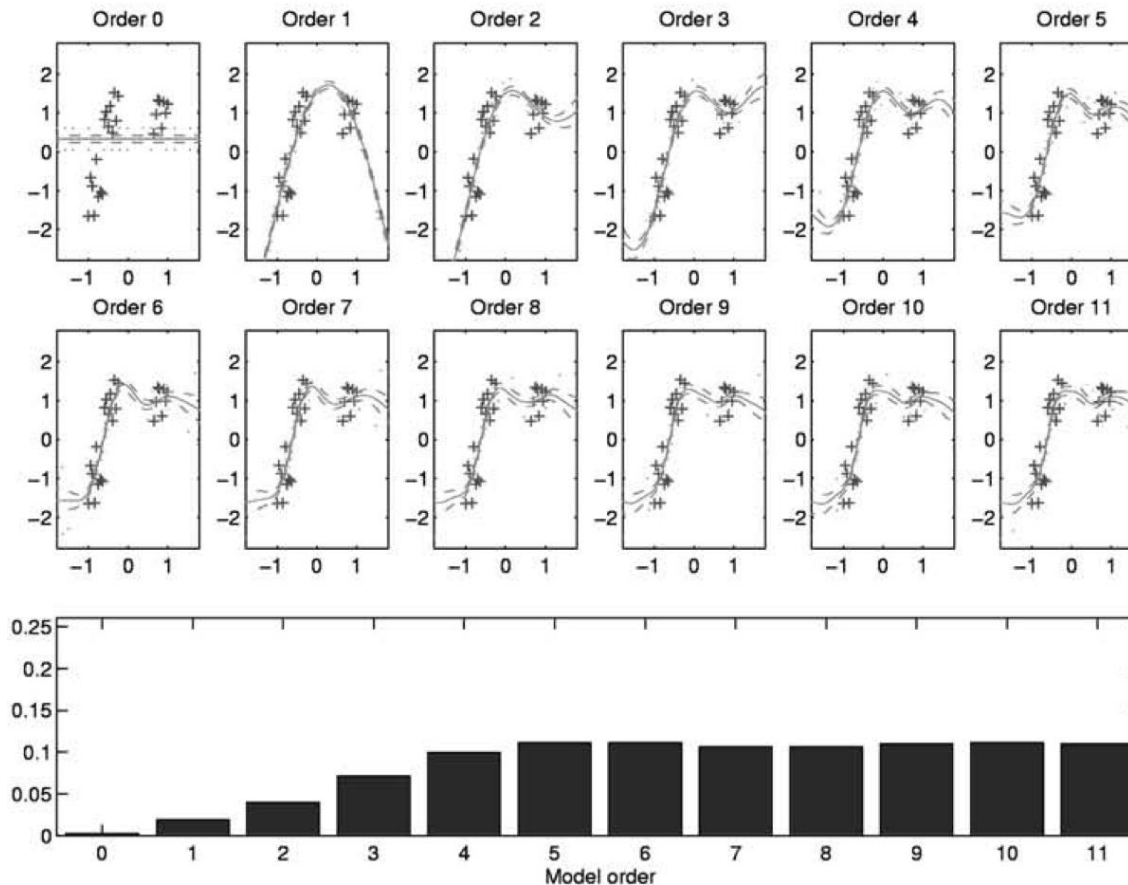
Gaussian prior with
variance independent of d



Penalizing Complexity



- Consider **trigonometric regression** as follows:
- $$y(x) = a_0 + \sum_{d=1}^D a_d \sin(dx) + b_d \cos(dx),$$



Gaussian prior with
precision proportional to d^3

Also, think about Gaussian
processes approximated with
random Fourier features
(frequencies decay fast)

Is Penalization the Answer? Or Just Over-Parameterization?



- Imposing **penalization on complexity** makes Bayesian model selection compatible with scaling laws
- However, scaling laws emerge even **without** this type of regularization*
- Is there any more to say about **priors****?
- Does this imply that in overparameterized models simplicity **emerges** over complexity***? Simplicity bias!

* except for stochastic optimization / weight decay / dropout

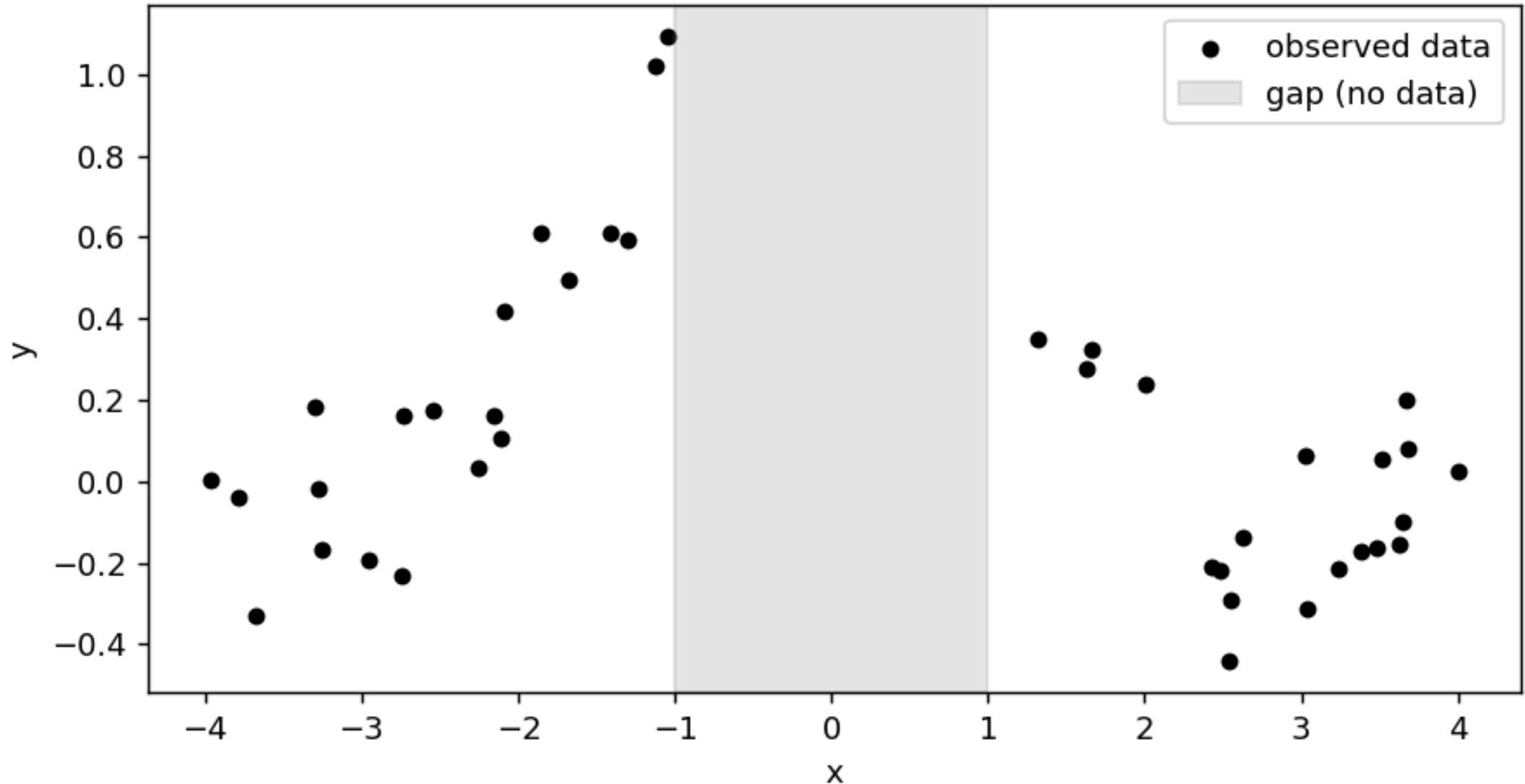
** Polson and Sokolov, arXiv 2025, Riha et al., arXiv 2026

*** Mingard et al., arXiv 2019l – Wilson, arXiv, 2025

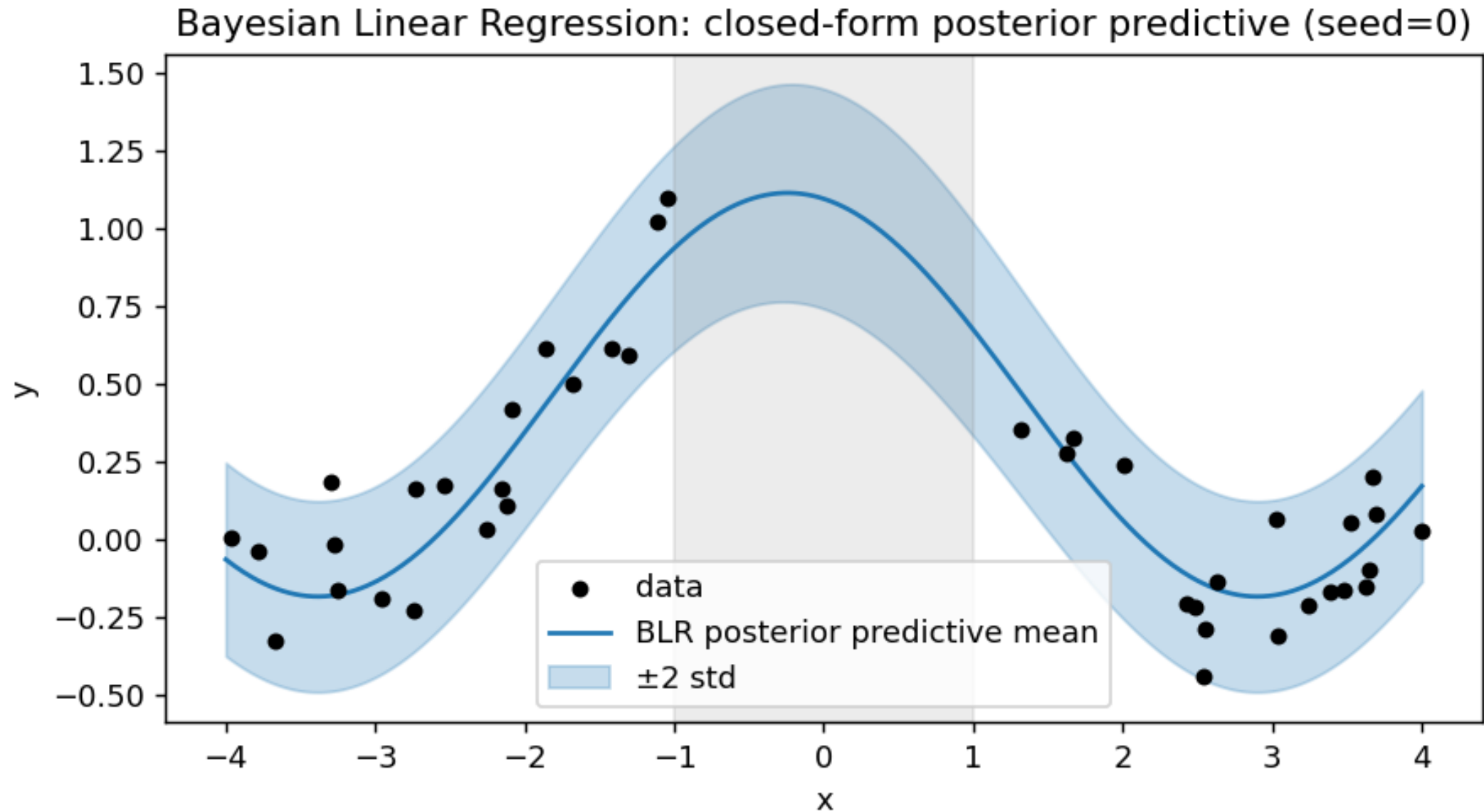
Example: Regression with Neural Networks



Dataset: simple trig-basis linear model + gap (seed=0)



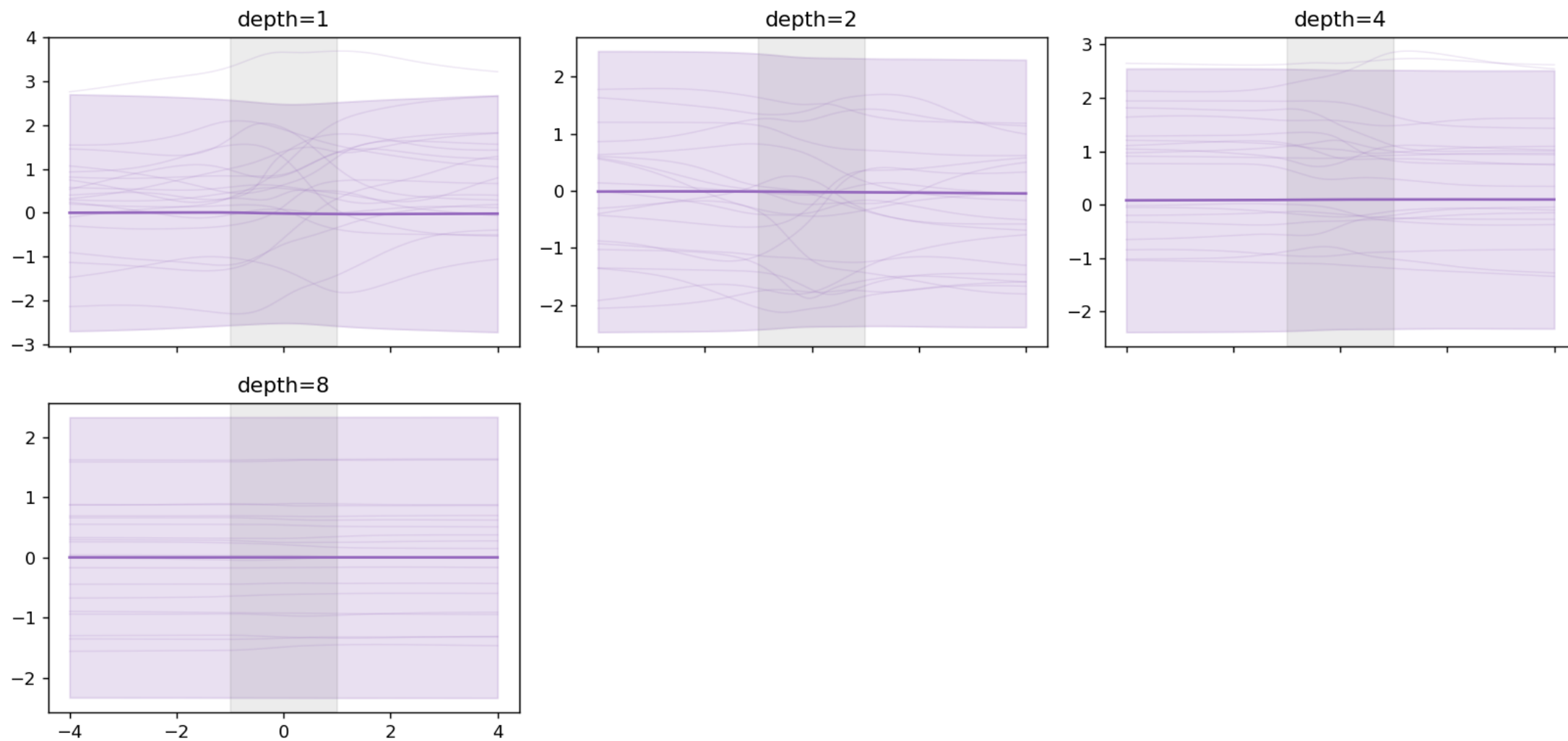
Solution with Correct Model Specification



Prior Predictive Distributions



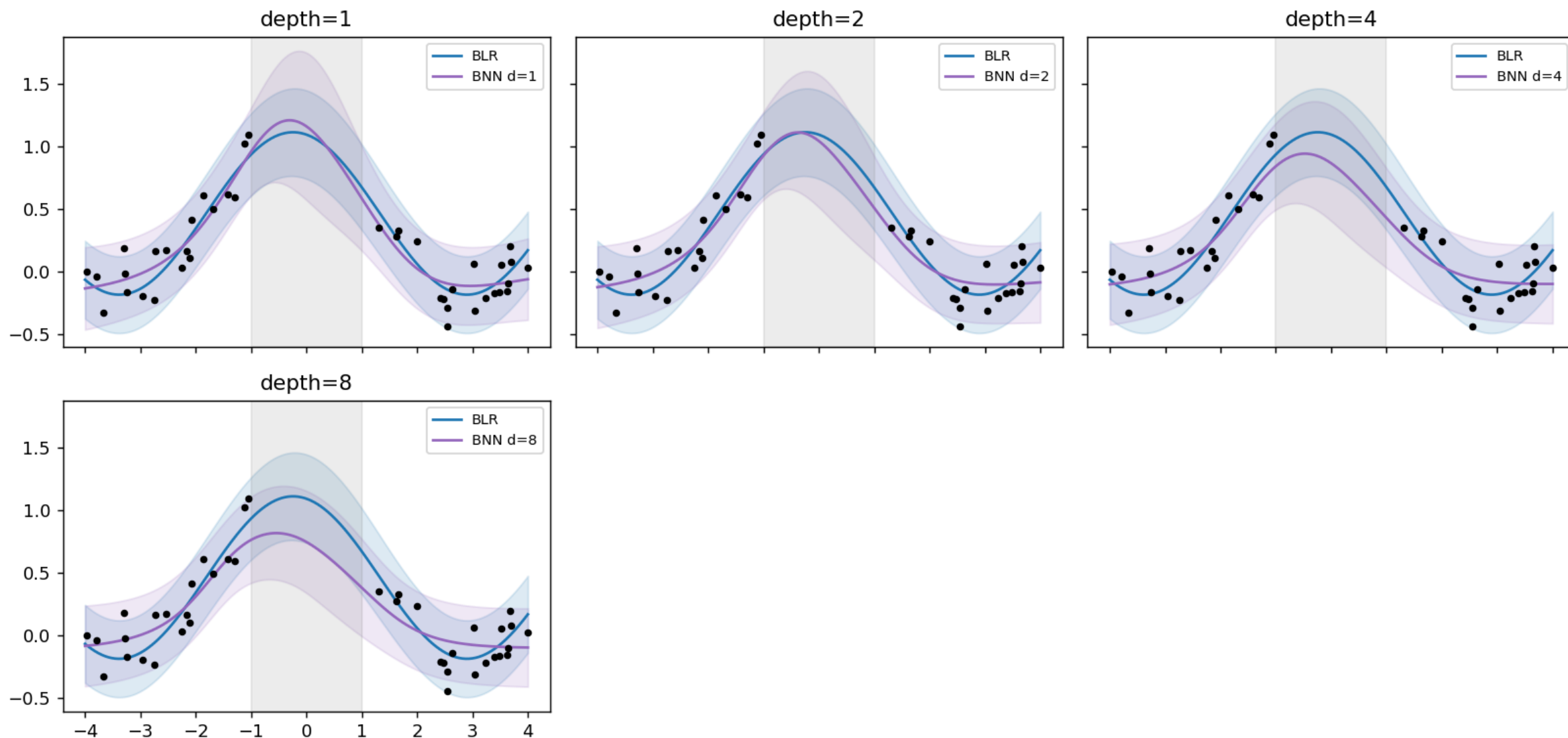
BNN prior predictive ($1/\sqrt{\text{fan_in}}$ -scaled prior, tanh activation, no data) across depths



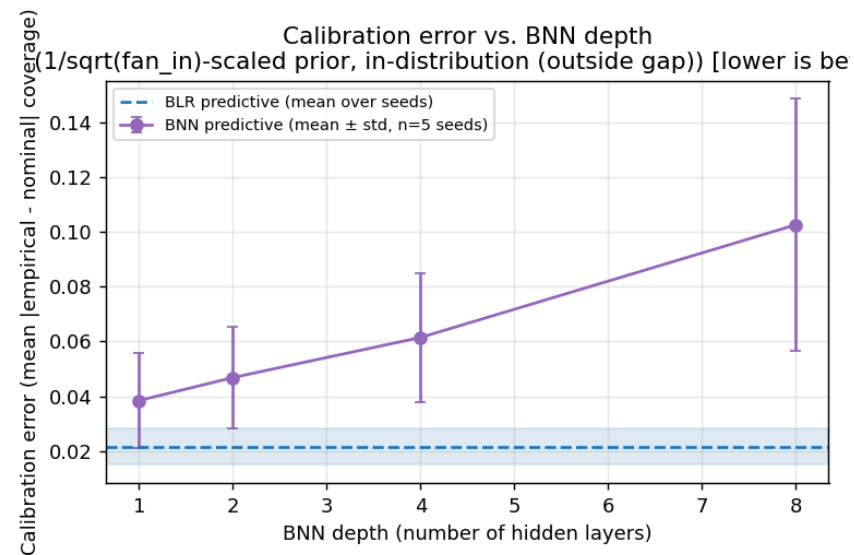
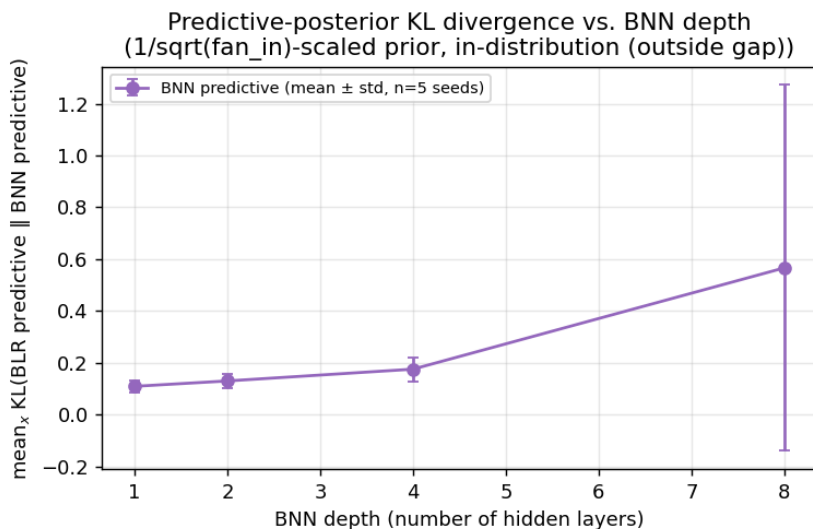
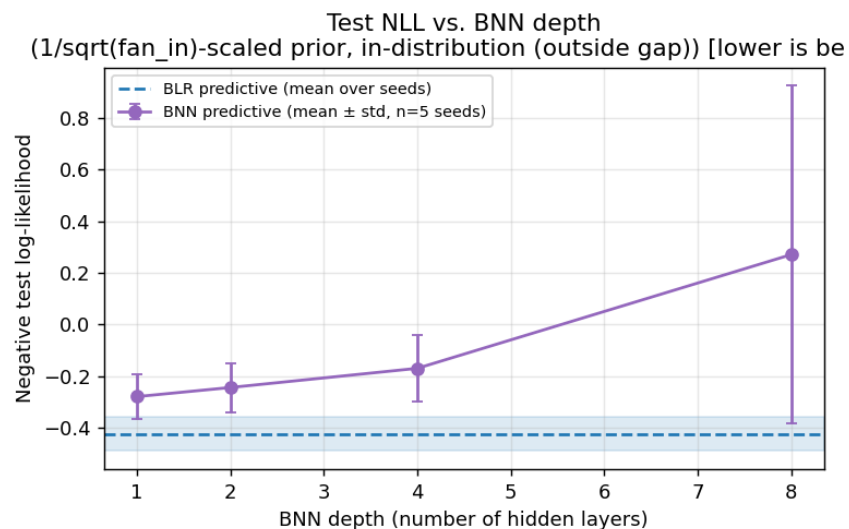
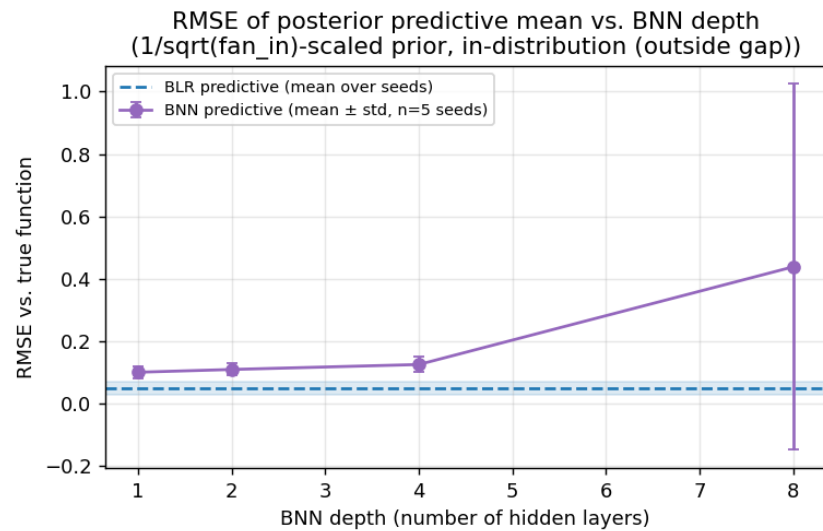
Example: Regression with Neural Networks



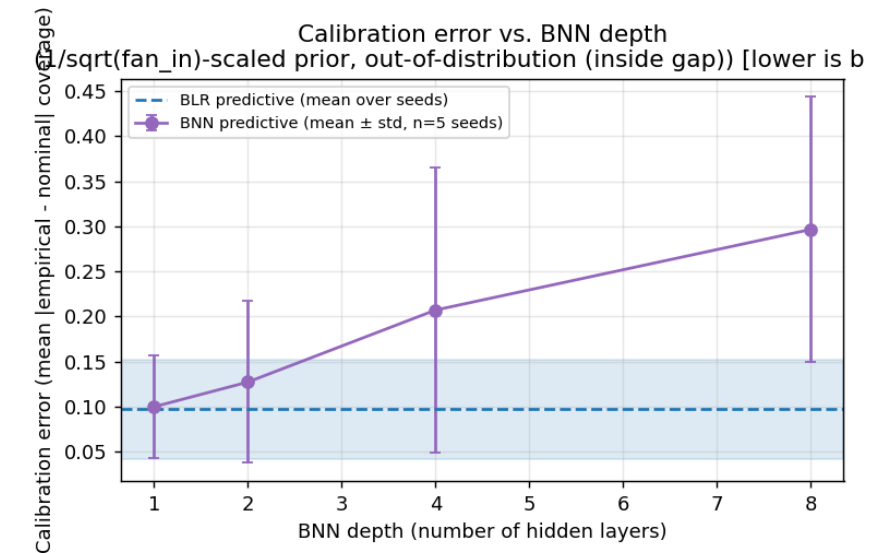
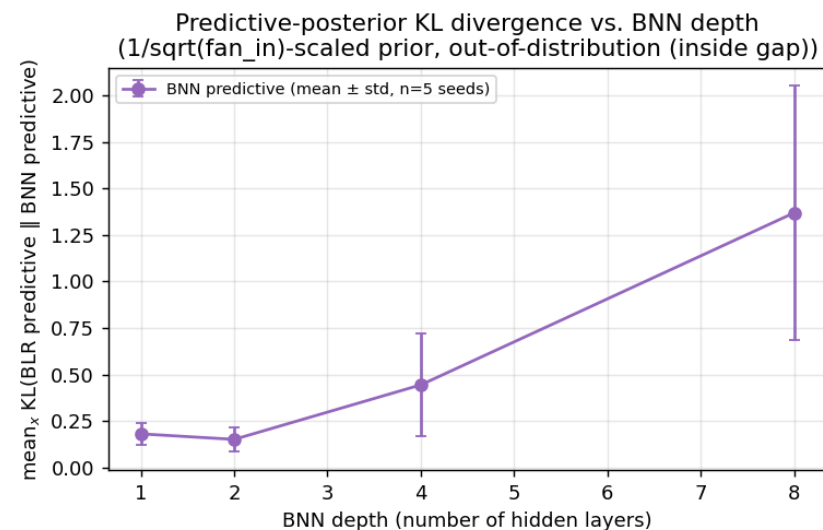
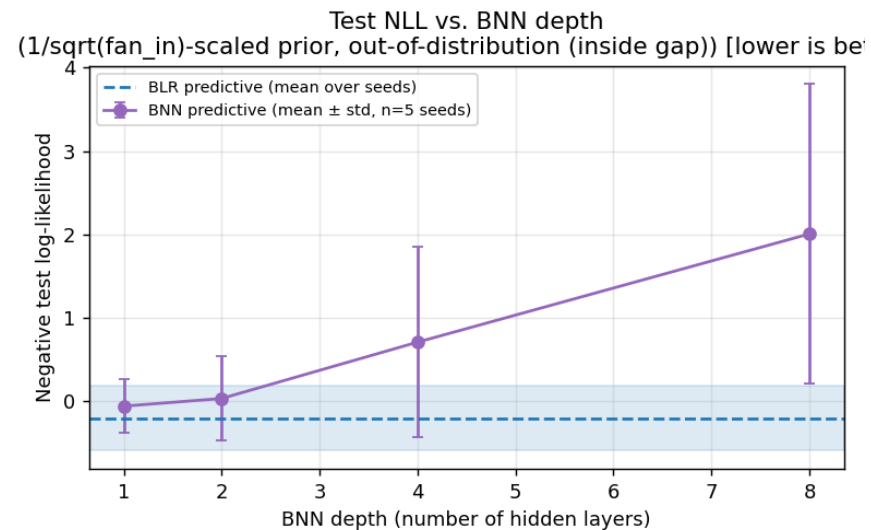
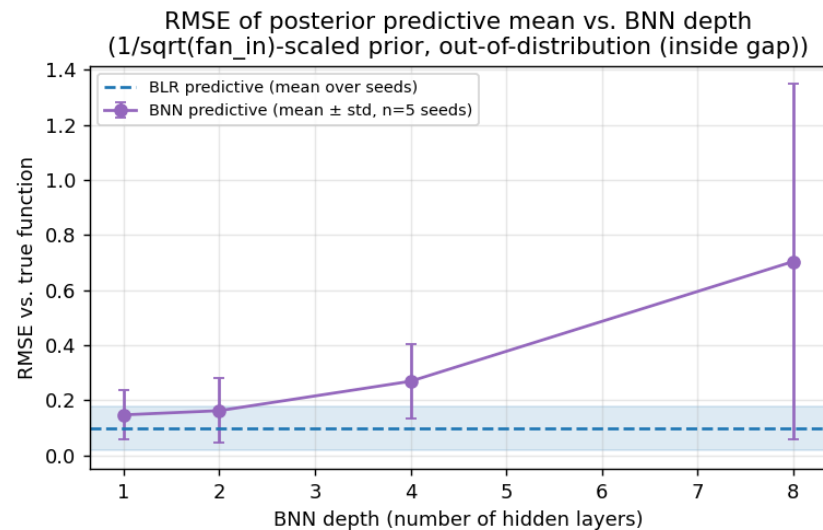
BLR (closed form) vs BNN ($1/\sqrt{\text{fan_in}}$ -scaled prior, tanh act.) posterior predictive (seed=0)



Performance In-Distribution



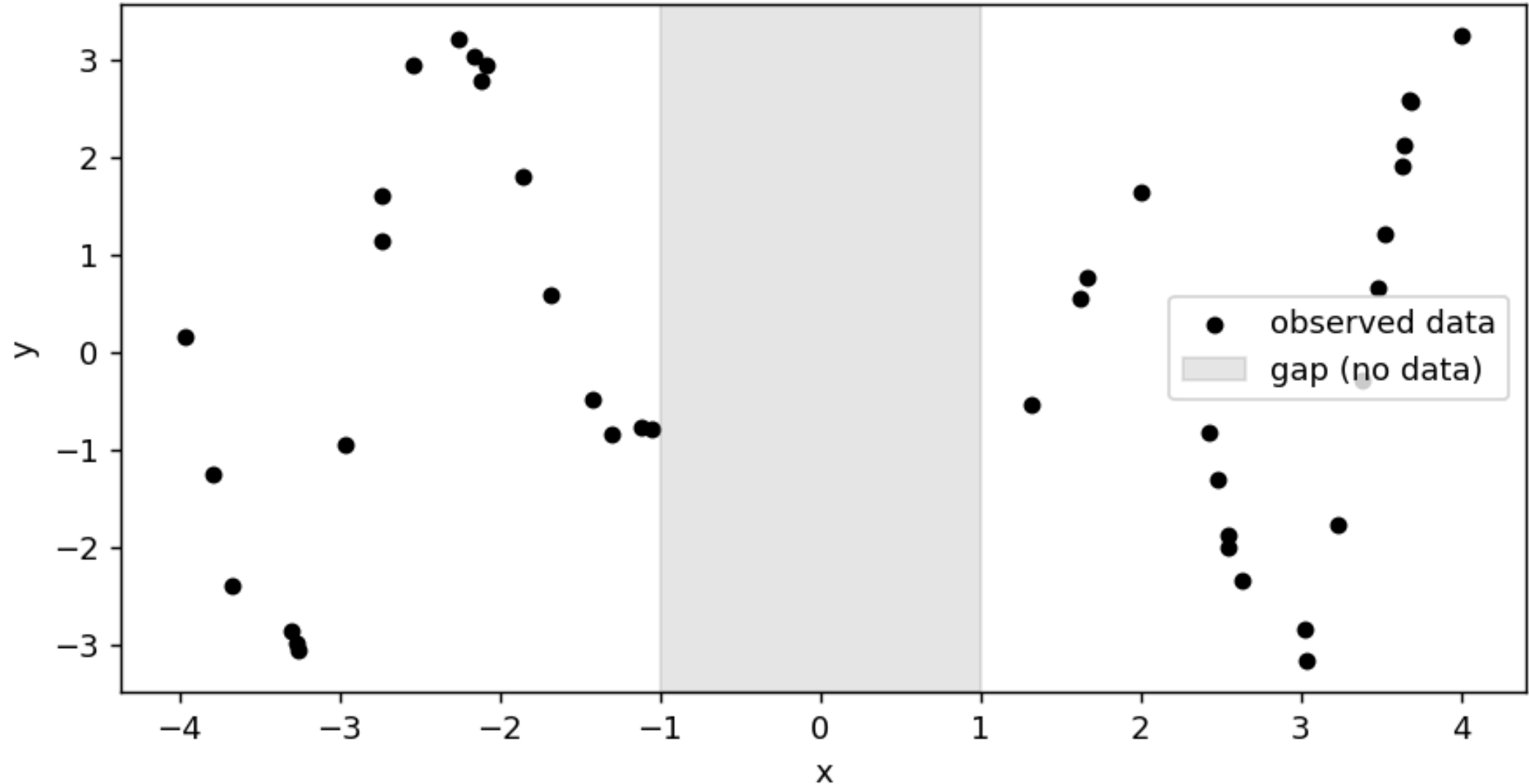
Performance Out-of-Distribution



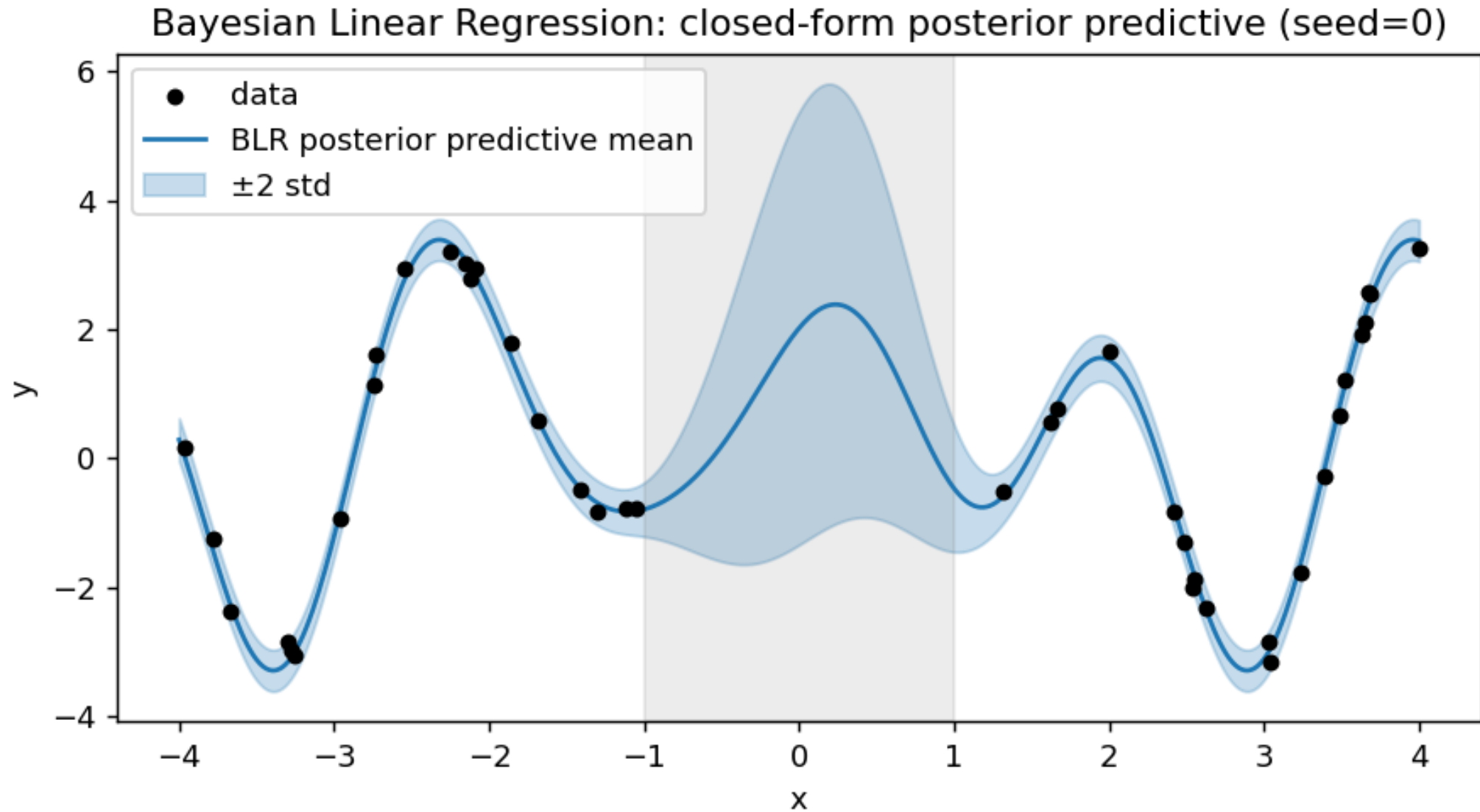
Example: Regression with Neural Networks



Dataset: simple trig-basis linear model + gap (seed=0)



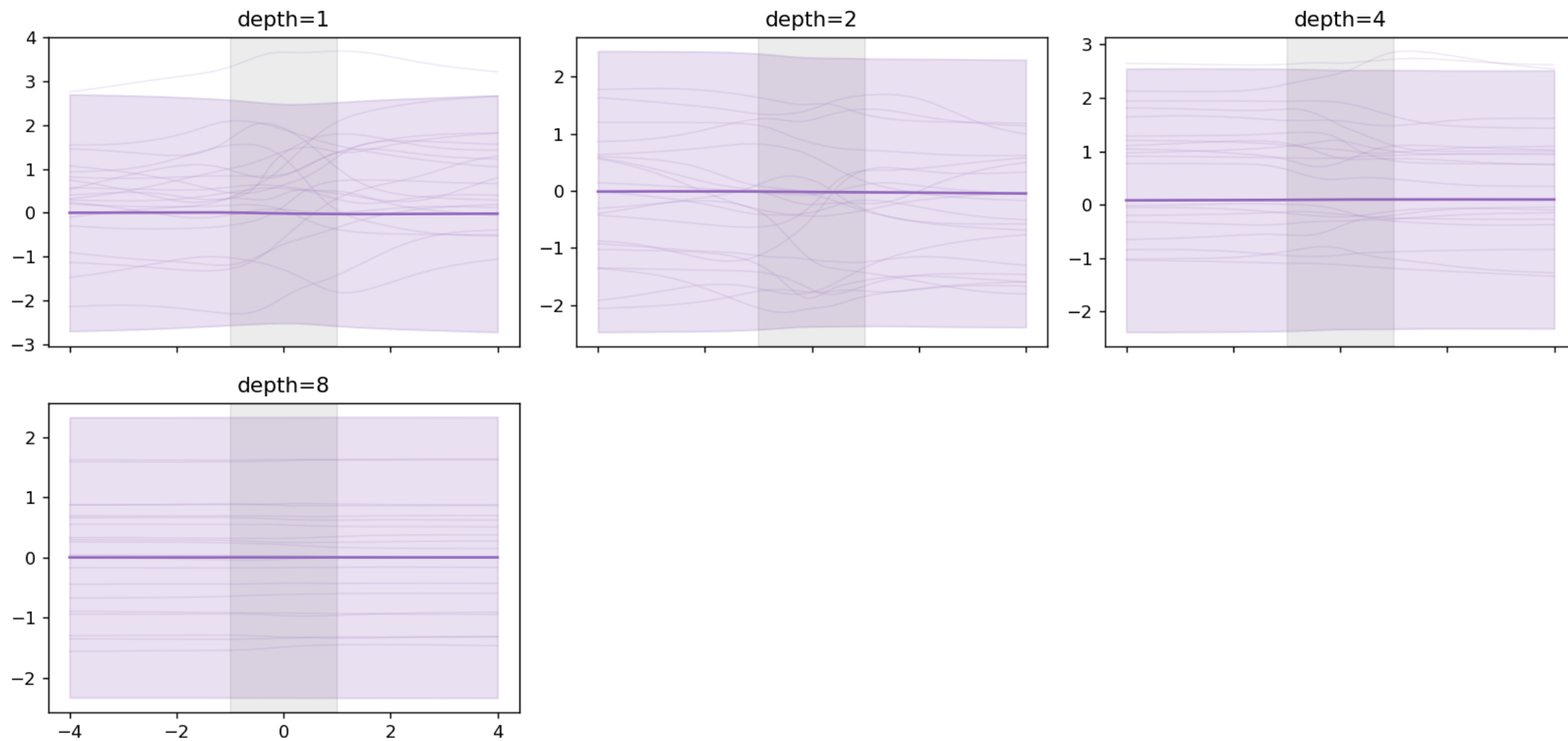
Solution with Correct Model Specification



Prior Predictive Distributions



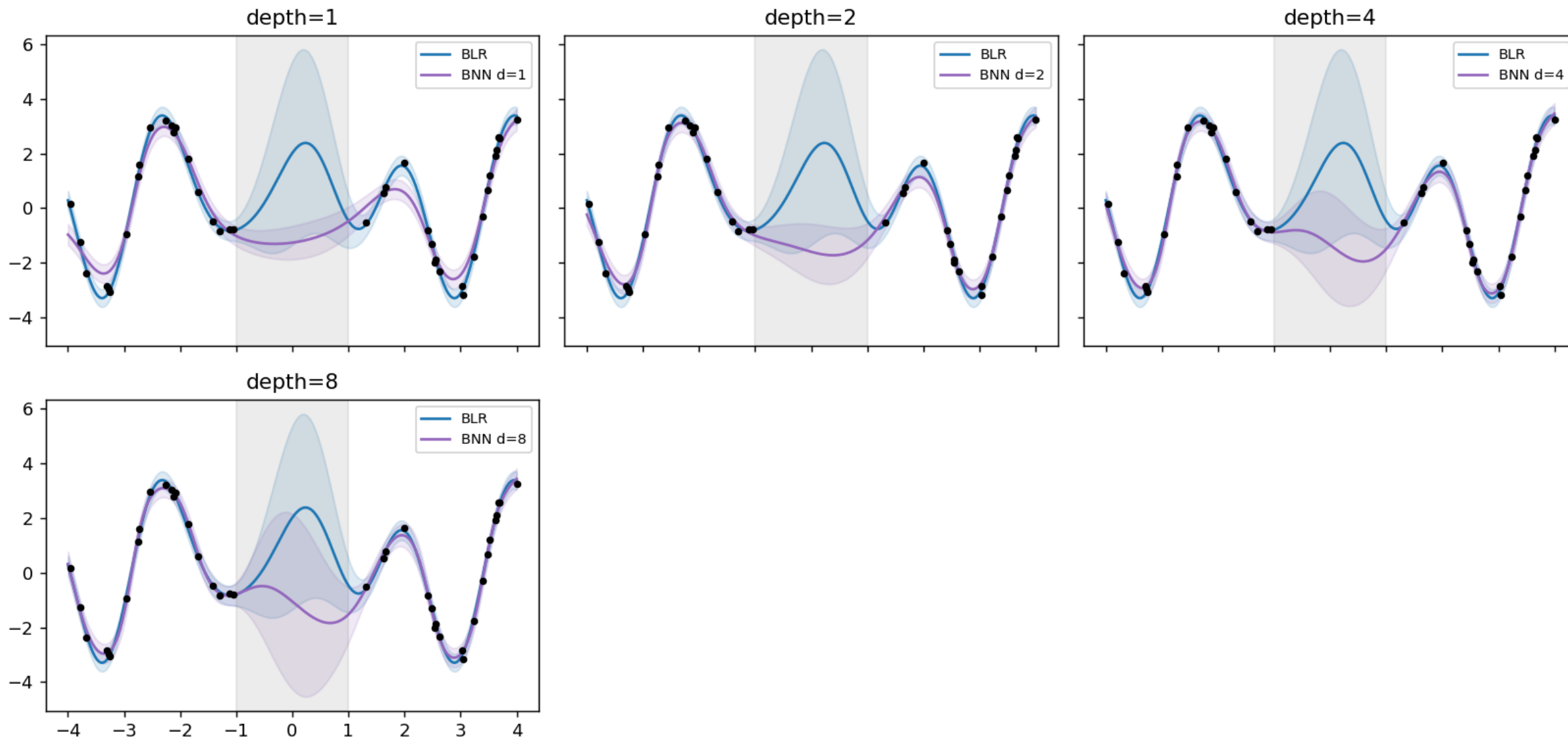
BNN prior predictive ($1/\sqrt{\text{fan_in}}$ -scaled prior, tanh activation, no data) across depths



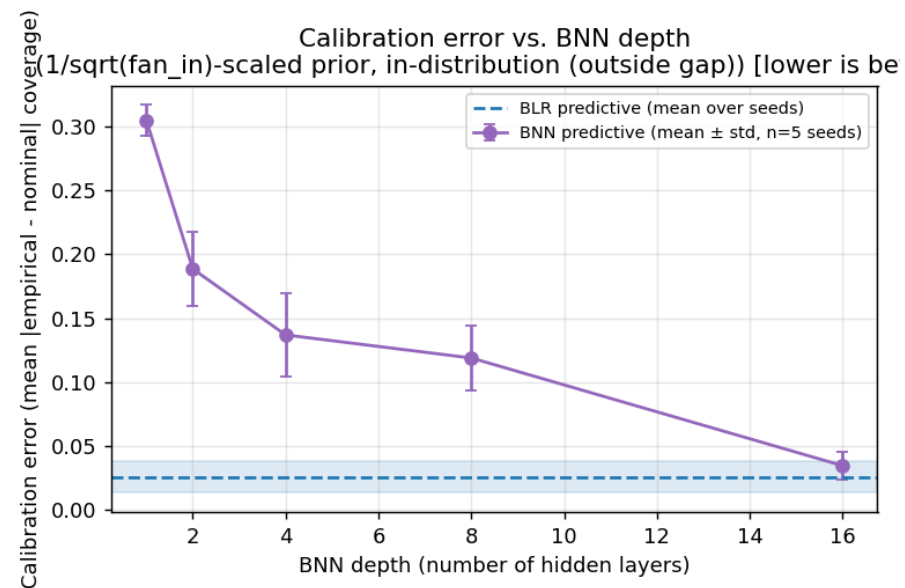
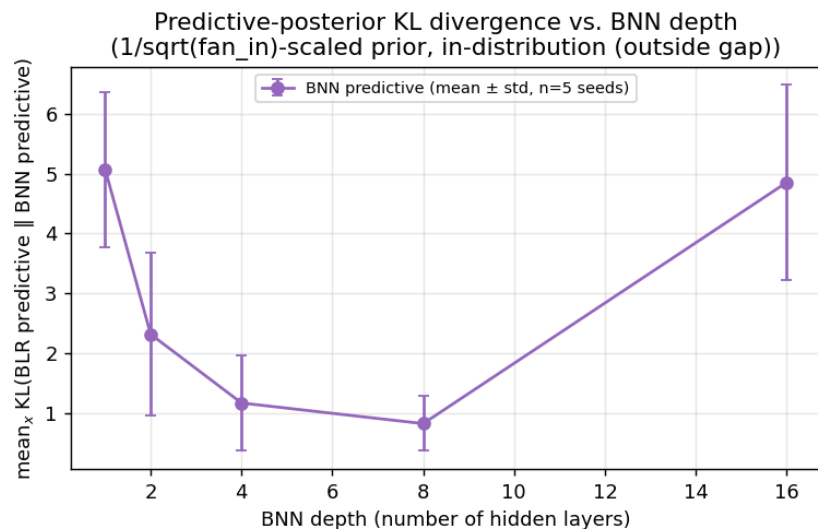
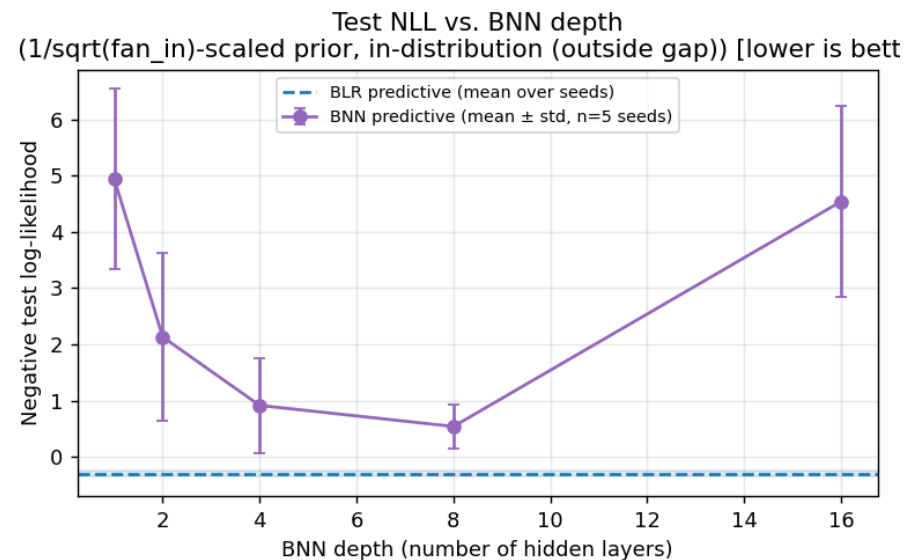
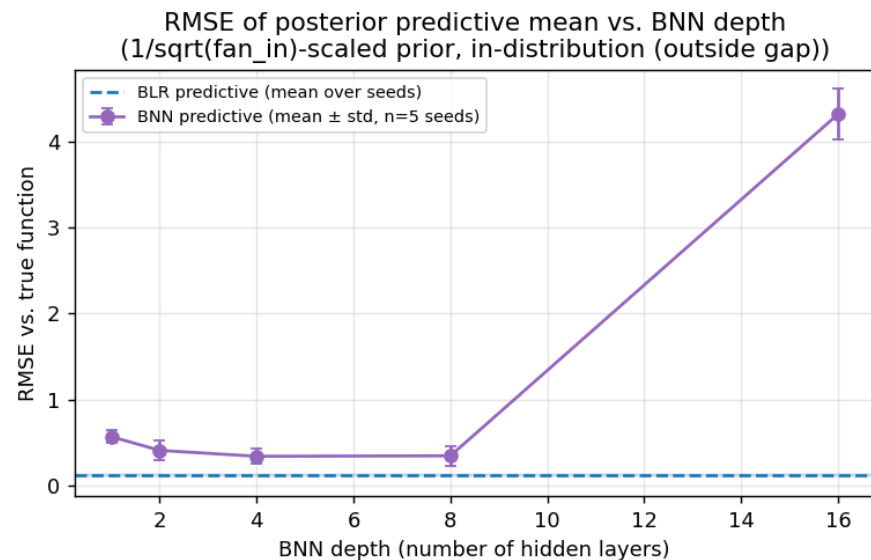
Example: Regression with Neural Networks



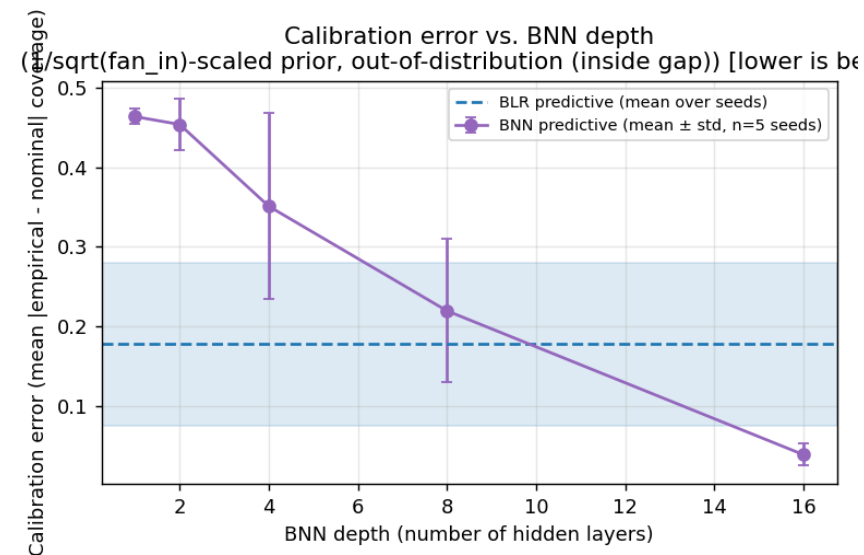
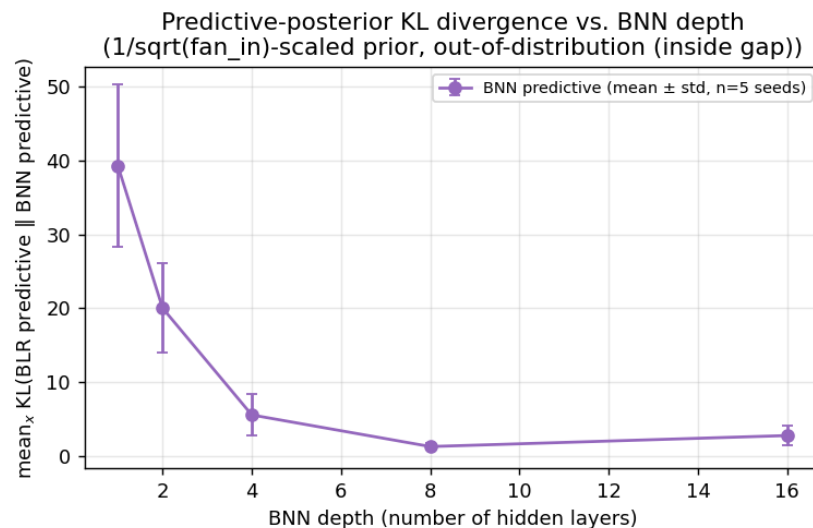
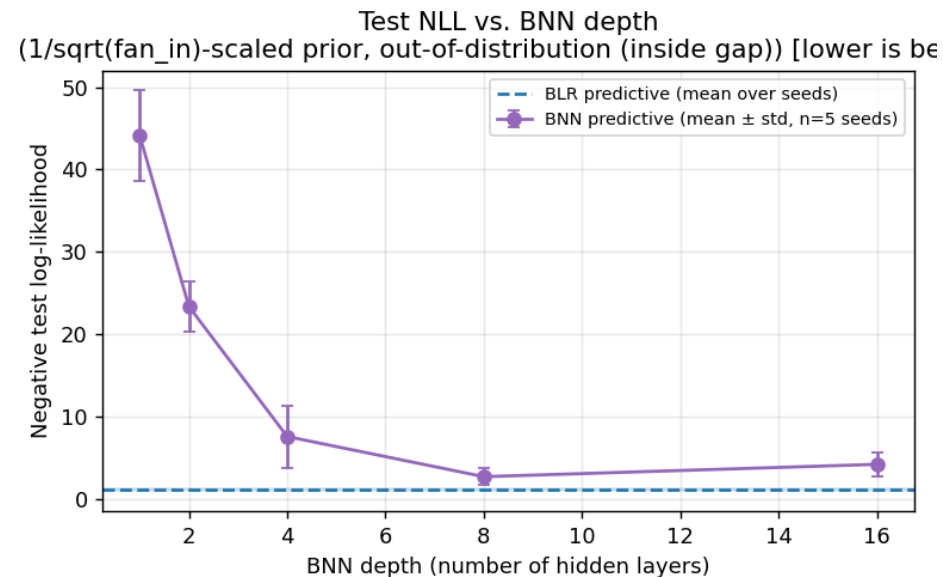
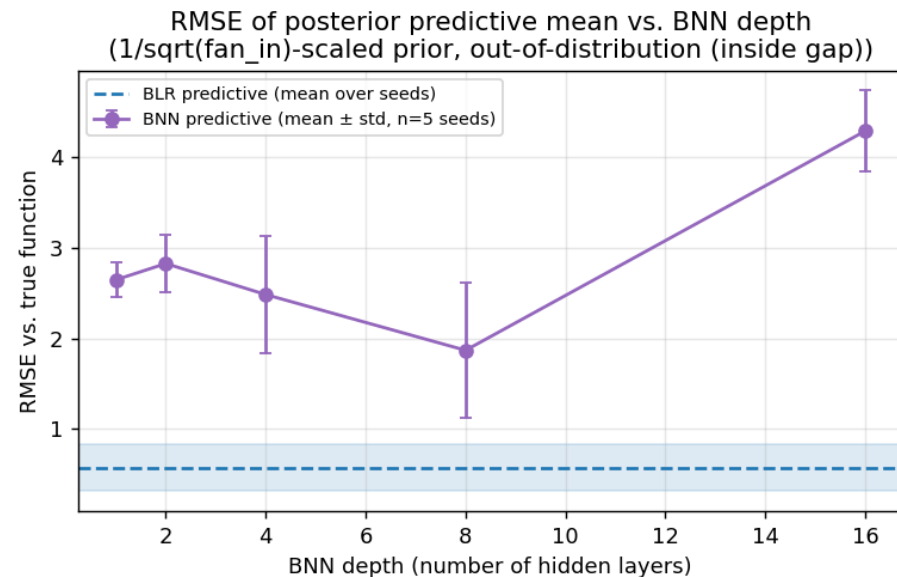
BLR (closed form) vs BNN ($1/\sqrt{\text{fan_in}}$ -scaled prior, tanh act.) posterior predictive (seed=0)



Performance In-Distribution



Performance Out-of-Distribution



Concluding Thoughts



- Scaling Laws are a fascinating phenomenon characterizing overparameterized models
- Bayesian model selection is “compatible” with scaling laws if we introduce complexity penalties
- But what if we don’t?
- Isn’t it still relevant to look for simpler and cheaper models, as opposed to giant and expensive ones?
- How to perform model selection in a practical way?

Model Selection for Bayesian Autoencoders

Ba-Hien Tran
EURECOM
(France)

Simone Rossi
EURECOM
(France)

Dimitrios Milios
EURECOM
(France)

Pietro Michiardi
EURECOM
(France)

Edwin V. Bonilla
CSIRO's Data61
The Australian National University
The University of Sydney
(Australia)

Maurizio Filippone
EURECOM
(France)

Abstract

We develop a novel method for carrying out model selection for Bayesian autoen-

Some More References



➤ Bayesian principles to understand and improve deep learning

Understanding and Improving GANs through Partial Stochasticity in Bayesian Neural Networks

Maurizio Filippone
Statistics Program
KAUST, Saudi Arabia
maurizio.filippone@kaust.edu.sa

Marius P. Linhard
Department of Computer Science
University of Verona, Italy
m.linhard@icloud.com

Abstract

Despite their challenging optimization, Generative Adversarial Networks (GANs) are popular and successful generative models. In this work, we explain their success and limitations by casting them as Bayesian neural networks with partial stochasticity. This interpretation allows us to rewrite the adversarial-style optimization of several variants of GANs as the optimization of a proxy for the likelihood obtained by marginalizing out stochastic variables, and to establish conditions of universal approximation. Most importantly, our unifying view also exposes a major statistical limitation of GANs in the lack of regularization for the generator. Results obtained on a wide range of GAN objectives and architectures indicate that regularizing the generator leads to consistent performance improvements, even in highly-engineered state-of-the-art GANs. In all, our work provides novel insights on how to improve GANs and paves the way to a deeper understanding of their theoretical foundations.

Published as a conference paper at ICLR 2026

OPTIMIZING DATA AUGMENTATION THROUGH BAYESIAN MODEL SELECTION

Madi Matymov*
KAUST
Saudi Arabia

Ba-Hien Tran
Huawei Paris Research Center
France

Michael Kampffmeyer
UiT The Arctic University of Norway
Norwegian Computing Center, Norway

Markus Heinonen
Aalto University
Finland

Maurizio Filippone
KAUST
Saudi Arabia

ABSTRACT

Data Augmentation (DA) has become an essential tool to improve robustness and generalization of modern machine learning. However, when deciding on DA strategies it is critical to choose parameters carefully, and this can be a daunting task which is traditionally left to trial-and-error or expensive optimization based on validation performance. In this paper, we counter these limitations by proposing a novel framework for optimizing DA. In particular, we take a probabilistic view of DA, which leads to the interpretation of augmentation parameters as model (hyper)-parameters, and the optimization of the marginal likelihood with respect to these parameters as a Bayesian model selection problem. Due to its intractability, we derive a tractable Evidence Lower BOund (ELBO), which allows us to optimize augmentation parameters jointly with model parameters. We provide extensive theoretical results on variational approximation quality, generalization guarantees, invariance properties, and connections to empirical Bayes. Through experiments on computer vision and NLP tasks, we show that our approach improves calibration and yields robust performance over fixed or no augmentation. Our work provides a